

TRHJ-1060
X-Band Four-Channel
Phased Array Chip

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Product Features

Operating frequency: 7 GHz–13 GHz (transmitting)

7 GHz–13 GHz (receiving)

Number of Rx/Tx channels: 4 channels each

Rx/Tx switching mode: half-duplex

Tx output power (OP1dB): 18.5 dBm

Tx link gain: 2 dB

Rx link gain: 0 dB

Phase modulation range/accuracy: 360°/5.625°

Amplitude modulation range/accuracy: 15.5 dB/0.5 dB

TX static power consumption: 0.3 W (single channel enabled)

RX static power consumption: 0.08 W (single channel enabled)

Packaging form: Wirebonding

Integrated temperature sensor, four-wire SPI, and on-chip ID bit

Application

Satellite communication

Overview

Wirebonding packaging is adopted for the chip. The working temperature range is -40°C to $+85^{\circ}\text{C}$. The chip is an mmWave low-power-consumption silicon-based (CMOS) chip with integrated 4 Rx/Tx channels and transmitting and receiving frequencies of 7 GHz–13 GHz. The chip is mainly used in phased array systems and paired with the corresponding gallium arsenide front-end chip. It can be used as a driver-class vector modulation. The chip enables Rx-Tx switches in half-duplex mode. In Tx mode, RF input signals enter into four Tx channels after passing through a power divider; in Rx mode, RF signals pass through four Rx channels and are combined into one way through a combiner before going to output. In the above two modes, the chip provides 6-bit phase adjustment (adjustment range: 360° , step: 5.625°) and 5-bit amplitude adjustment (adjustment range: 15.5dB, step: 0.5dB).

The chip uses a simple four-wire serial port interface (SPI) for register control.

Packaging form: Wirebonding, dimension: $6640*5050*0.2$ (mm)

Functional Block Diagram

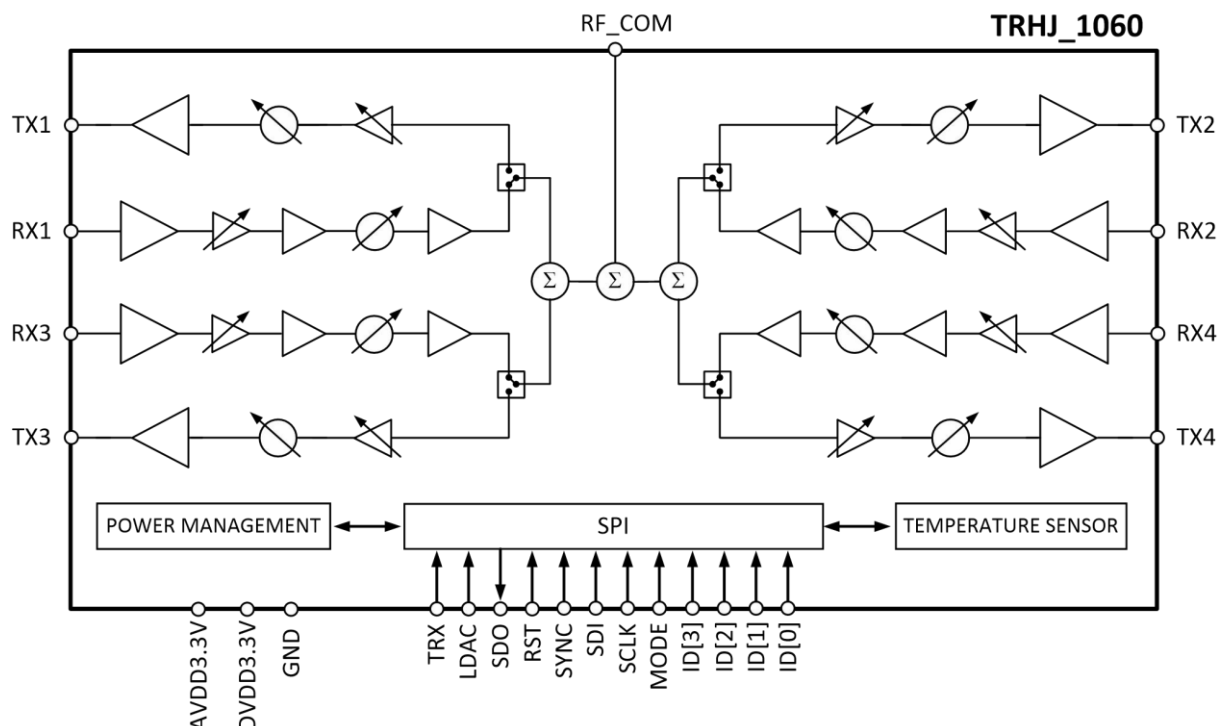


Figure 1 Functional Block Diagram of Chip

Technical Specifications

Unless otherwise specified, the chip is set to the maximum channel gain and standard bias conditions in the normal working state, and all bias register values are set to the default state. AVDD=3.3V, DVDD=3.3V, TA=25°C.

Table 1 Technical Specifications of Chip

S/N	Feature	Conditions: Unless otherwise specified, -55℃≤TA≤85℃, operating frequency: 7–13 GHz, operating voltage: digital voltage (3.3 V), analog voltage (3.3 V)	Test Indicator		Unit
			Minimum Value	Maximum Value	
Tx channel					
1	Tx gain (Single channel test, other ports in the load state)	TA=25℃, including power divider loss	0	4	dB
2	Output P1 dB	TA=25℃	18.5	20	dBm
3	Phase shift accuracy (RMS)	TA=25℃, phase shift bit: 6 bits, step: 5.625°	1.2	2.5	°
4	Additional attenuation of phase shift	TA=25℃	-0.5	0.5	dB
5	Attenuation control accuracy (RMS)	TA=25℃, attenuation bit: 5 bits, step: 0.5 dB	0.1	0.7	dB
6	Additional phase shift of attenuation	TA=25℃	0	7	°
7	Input/output standing wave ratio	TA=25℃, input standing wave	1.16	2.4	
		TA=25℃, output standing wave	1.03	2.0	
8	In-band gain fluctuation	Ground state, TA=25℃	-2	2	dB
9	Tx 3.3V current	TA=25℃, ground state, single-channel dynamic current	/	92	mA
Rx channel					
10	Rx gain (Single channel test, other ports in the load state)	TA=25℃, including power divider loss	-1	2	dB
11	Noise figure	TA=25℃, including 1-to-4 power divider loss	/	/	dB
12	Input P1dB	TA=25℃, ground state (zero amplitude modulation and phase modulation)	-0.5	2	dBm
13	Phase shift accuracy (RMS)	TA=25℃, phase shift bit: 6 bits, step: 5.625°	1	2.5	°
14	Additional attenuation of phase shift	TA=25℃	-0.5	0.5	dB
15	Attenuation control accuracy (RMS)	TA=25℃, attenuation bit: 5 bits, step: 0.5 dB	0.2	0.4	dB
	Additional phase shift of attenuation	TA=25℃	0	4	°
16	Input/output standing wave ratio	TA=25℃, input standing wave	1.03	1.4	
17		TA=25℃, output standing wave	1.16	2.4	
18	In-band gain fluctuation	Ground state, TA=25℃	-1.5	1.5	dB
19	Rx 3.3V current	TA=25℃, ground state, four-channel dynamic current	/	70	mA

Absolute Maximum Rating

Table 2. Absolute Maximum Rating Value

Parameter	Absolute Maximum Rating	Unit
AVDD	3.6	V
DVDD	3.6	V
Operating case temperature range	-40°C to +85°C	°C
Storage temperature range	-55°C to +125°C	°C
Maximum junction temperature	135	°C
Pin temperature (welding, 60 seconds)	260	°C
Electrostatic discharge sensitivity (ESD)	1000	V

Note that exceeding or equal to the above absolute maximum rating may cause permanent damage to the product. This is only the rated value and does not mean that the device can work normally under these conditions or under any other conditions beyond the specifications shown in the operation section of the Technical Specification. Long-term operation beyond the maximum rated value will affect the reliability of the product.



ESD (electrostatic discharge) sensitive device

Charged devices and circuit boards may discharge without detection. Although this product has patented or proprietary protection circuit, the device may be damaged in case of high-energy ESD. Therefore, appropriate ESD precautions should be taken to avoid device performance degradation or function loss.

Typical Performance Parameters

Tx Mode

Unless otherwise specified, the maximum channel gain and standard bias conditions are set in the Rx state. AVDD=3.3V, DVDD=3.3V, TA=25°C.

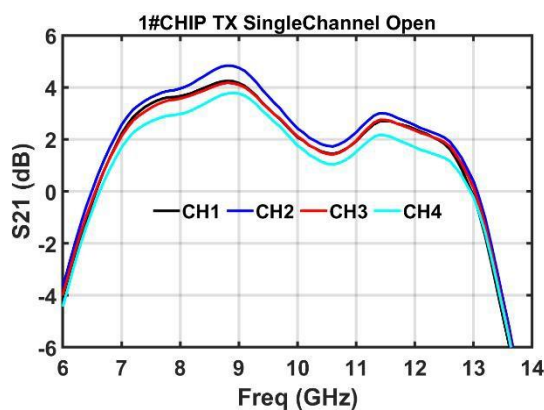


Figure 2. Relationship between Gain and Frequency in Different Channels

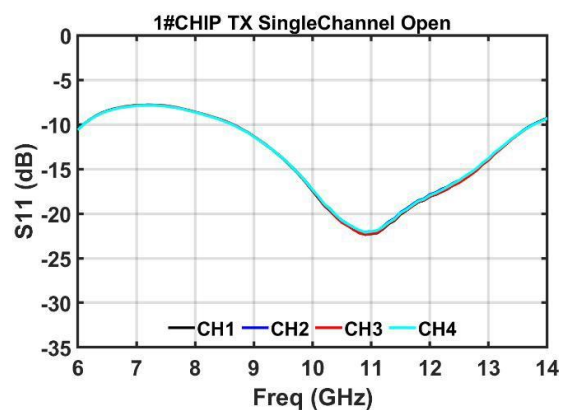


Figure 3. Input Reflection Coefficient

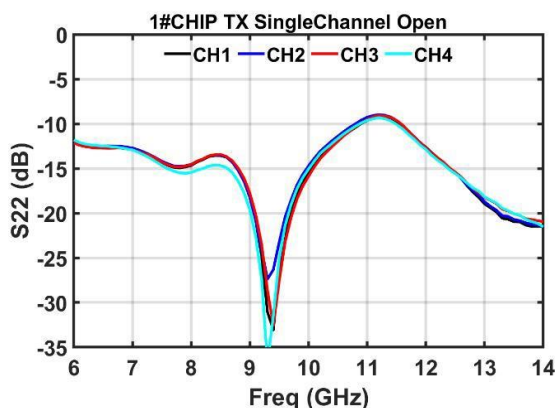


Figure 4. Output Reflection Coefficient

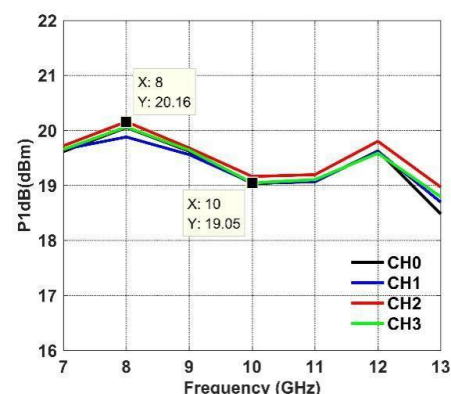


Figure 5. Output P1dB Power

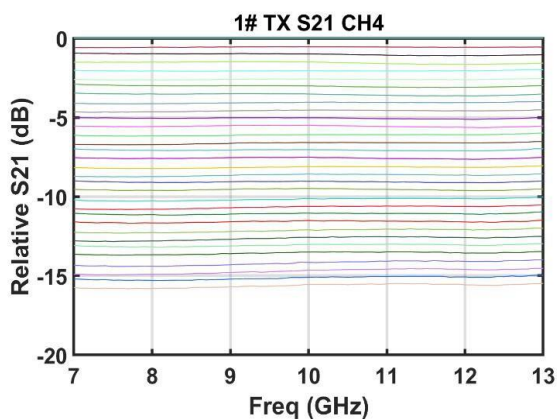


Figure 6. Relationship between Amplitude Modulation and Frequency

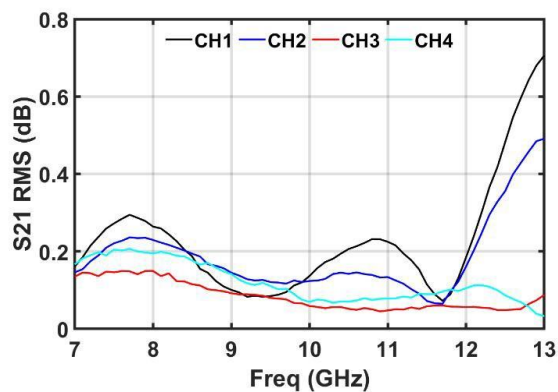


Figure 7. Relationship between AM RMS and Frequency in Different Channels

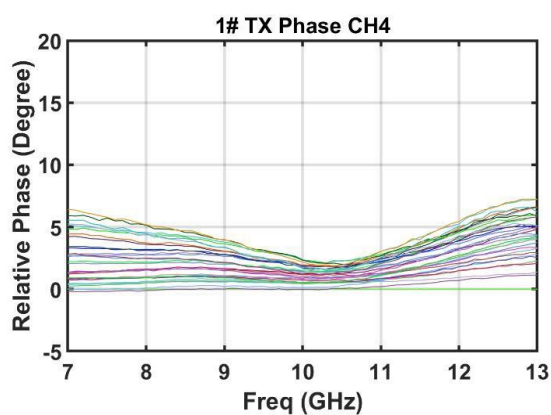


Figure 8. Curve of Amplitude Modulation Additional Phase Movement with Frequency

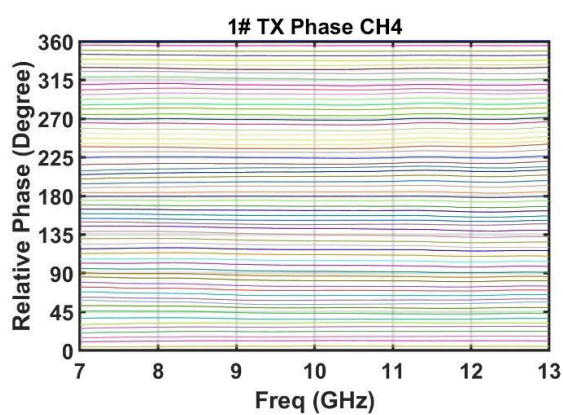


Figure 9. Relationship between Phase Modulation and Frequency

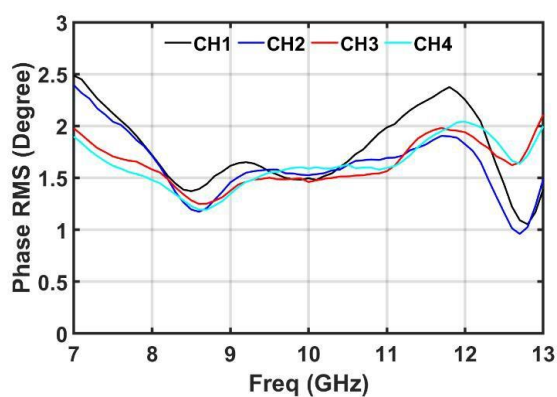


Figure 10. Relationship between PM RMS and Frequency in Different Channels

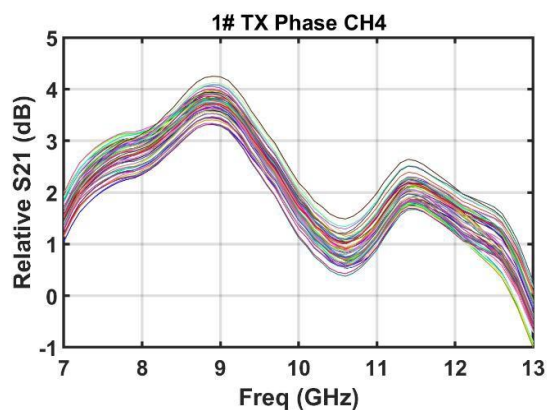


Figure 11. Curve of Phase Modulation Additional Attenuation Movement with Frequency

Rx Mode

Unless otherwise specified, the maximum channel gain and standard bias conditions are set in the Tx state. AVDD=3.3V, DVDD=3.3V, TA=25°C.

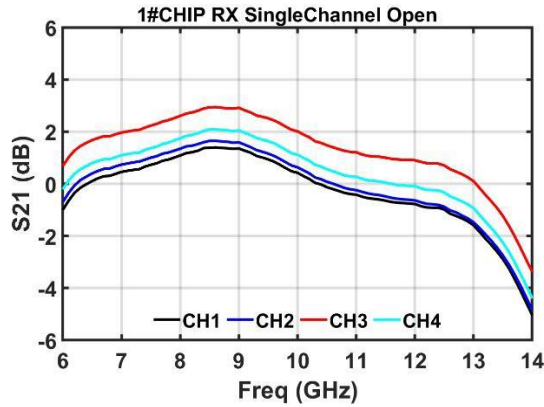


Figure 12. Relationship between Gain and Frequency in Different Channels

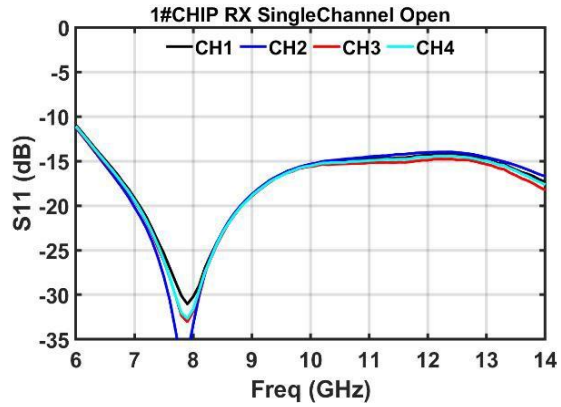


Figure 13. Input Reflection Coefficient

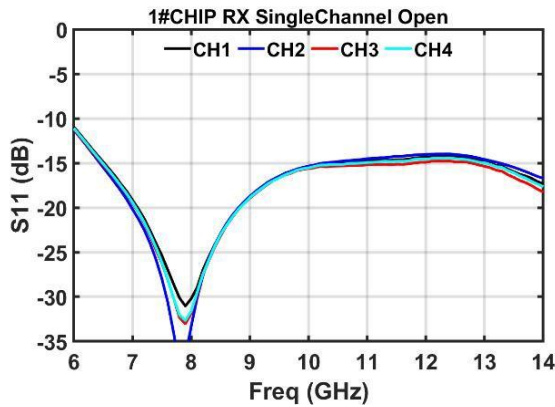


Figure 14. Output Reflection Coefficient

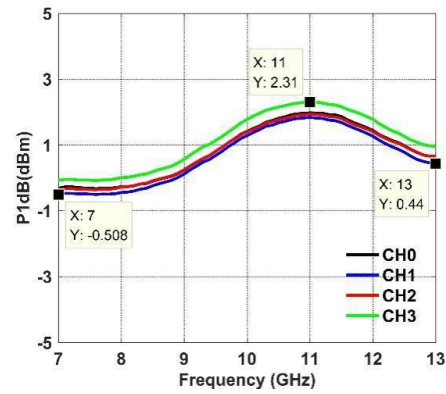


Figure 15. Input 1 dB Compression Power

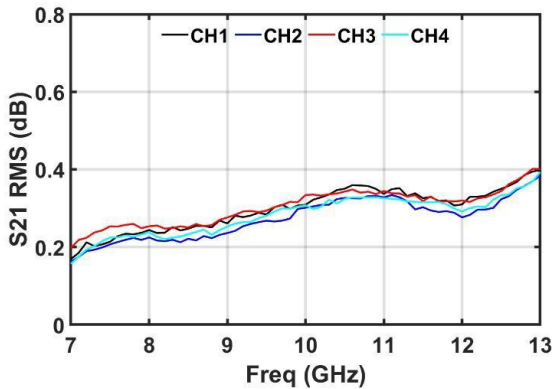


Figure 16. Relationship between Amplitude Modulation and Frequency

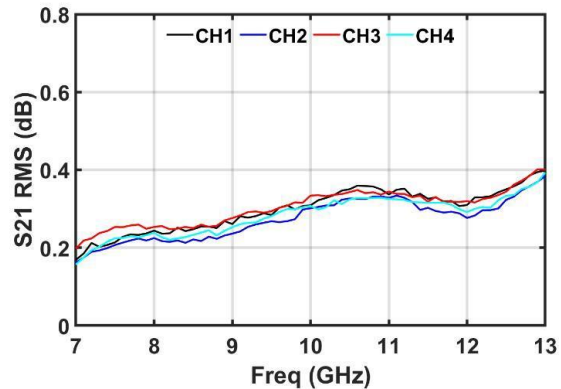


Figure 17. Relationship between AM RMS and Frequency in Different Channels

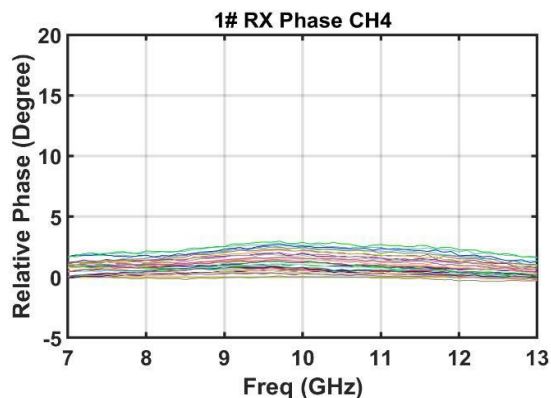


Figure 18. Curve of Amplitude Modulation Additional Phase Movement with Frequency

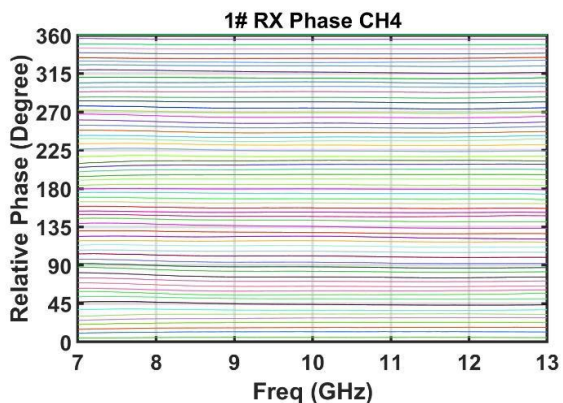


Figure 19. Relationship between Phase Modulation and Frequency

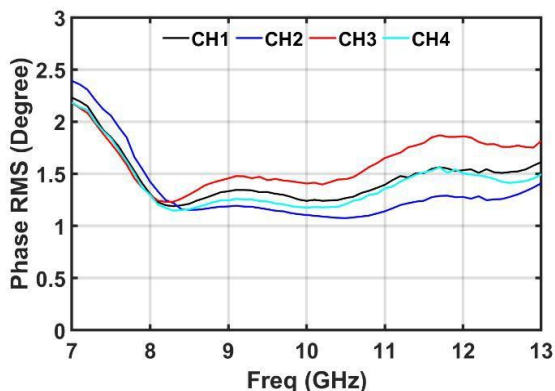


Figure 20. Curve of Phase Modulation Additional Attenuation Movement with Frequency

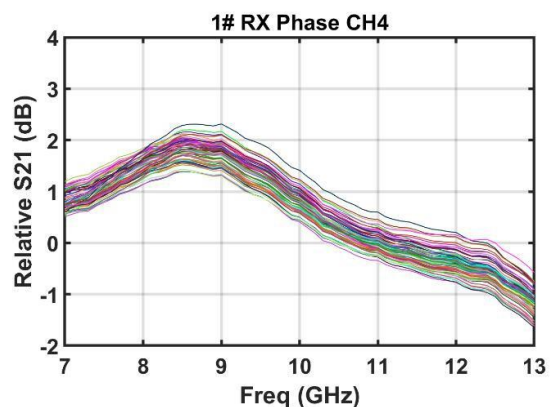


Figure 21. Curve of Phase Modulation Additional Attenuation Movement with Frequency

Chip Dimensions

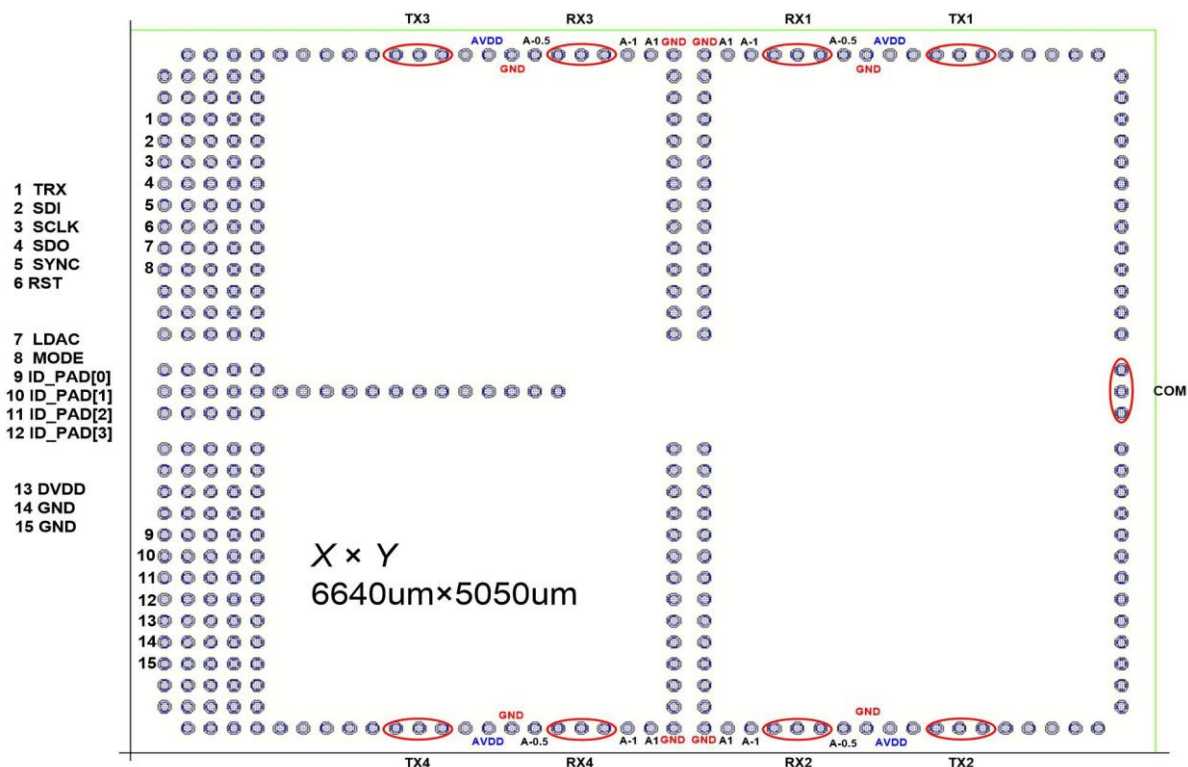


Table 3: Chip Size

Dimension Symbol	Value (unit: mm)		
	Minimum	Typical Value	Maximum
W	6.6	6.64	6.68
L	5.01	5.05	5.09
H	0.15	0.2	0.25

Remark: RF GSG is tested by an RF probe with a distance of 150 μm . The peripheral digital power supply of the chip is 3.3 V, and a 100pF filter capacitor is added. A 100pF decoupling capacitor is added to each of the four analog 3.3 V power supply voltage points.

Chip PAD Definition

Table 4: Size and Functions of the Bonding Area

Pressure Point	Symbol	Functional Description	Size (μm^2)
1	RX1-RX4	The first to the fourth RF Rx end	80×80
2	TX1-TX4	The first to the fourth RF Tx end	80×80
3	GND	Ground	80×80
4	AVDD	3.3 V analog circuit power supply	80×80
5	DVDD	3.3 V digital circuit power supply	80×80
6	TRX	Rx/Tx switching signal, high level for Tx, low level for Rx	80×80
7	SDO	Serial data output	80×80
8	SCLK	Input clock signal	80×80
9	SDI	Serial data input	80×80
10	SYNC	Chip select, enabled at the low level	80×80
11	LDAC	Latch signal, rising edge trigger	80×80
12	RST	Reset signal	80×80
13	MODE	Switching between ID mode and ID-free mode	80×80
14	ID[0]	ID bit	80×80
15	ID[1]	ID bit	80×80
16	ID[2]	ID bit	80×80
17	ID[3]	ID bit	80×80
18	A1	Add 1 dB bit to the gain of the first to the fourth channel (1 for each channel)	80×80
19	A-0.5	Reduce 0.5 dB bit from the gain of the first to the fourth channel (1 for each channel)	80×80
20	A-1	Reduce 1 dB bit from the gain of the first to the fourth channel (1 for each channel)	80×80

Note: The TTL high level of the digital input port is 2.8–4.5 V, and the low level is -0.5–0.5 V; the TTL high level of the digital output port is 2.8–3.3 V, and the low level is -0.5–0.5 V;

Chip Digital Control

Introduction to Digital Functions

The following sections introduce the working principle, communication protocol and usage of SPI digital interface in TRHJ-1060 chip.

SPI digital module needs DDVD and TTL (3.3 V) level. DVDD 3.3 V is used for the power supply of the IO module in SPI digital module. The chip has a built-in LDO from 3.3 V to 1 V for the power supply of the core module in SPI digital module. TTL (3.3 V) level is used for IO signals in SPI digital module.

The module has the following characteristics:

- Control the switch between Tx and Rx channels
- Control the power supply status of 4 Tx/Rx RF channels and support reading and writing
- Read and write 4 Tx/Rx RF channels with 6-bit amplitude and 6-bit phase
- Read and write calibration codes for 32-level amplitude and 64-level phase
- Support reading and writing up to 256 (actually 51 used) 8-bit on-chip control data (such as bias, etc.)
- Support readback of up to 8 12-bit on-chip signal data (such as temperature, etc.)
- Adopt pipelined read-write mode, with high throughput

SPI Interface Introduction

Table 5 Chip Interface Function

API Name	Function	Electrical Level Standard
TRX	Switch between Tx and Rx, and between Tx and Rx phase tables	TTL (3.3V)
LDAC	Load the amplitude and phase already written, and the falling edge is valid.	TTL (3.3V)
ID[3:0]	4-bit chip ID	TTL (3.3V)
MODE	Switching between ID mode and ID-free mode	TTL (3.3V)
SDO	Chip SPI output	TTL (3.3V)
RST	Reset signal, low-level effective (256 clock synchronous reset)	TTL (3.3V)
SYNC	SPI chip selection, low-level effective	TTL (3.3V)
SDI	Chip SPI input	TTL (3.3V)
SCLK	Chip SPI clock	TTL (3.3V)
DVDD	Digital power supply	3.3V (±10%)

MODE: the switch of chip ID MODE. When MODE is 1 (pulled up to 3.3 V), SPI works in ID state. After ID is correctly configured, it can support reading and writing. When MODE is 0 (grounded), SPI works in no ID state. At this time, the ID bit configuration is invalid. This state only supports write operation. (See “Chip ID Configuration” for ID bit configuration).

RST: reset signal, enabled at the low level. When it remains low for at least 128 clock cycles, it is pulled high and SPI is reset. To ensure the normal operation of the chip, reset operation should be performed first after the chip is powered on.

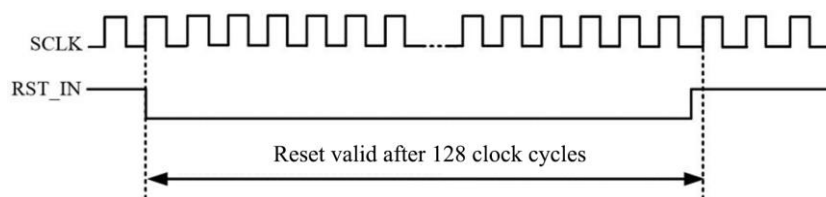


Figure 22 SPI Reset Time Sequence

TRX: responsible for switching between Tx and Rx, and between Tx and Rx phase tables. If TRX=1, Tx mode; if TRX=0, Rx mode.

SPI Hardware Topology

The topological structure of the SPI hardware is shown in Figure 23. The RF front-end chip is equipped with a 4-bit chip ID (ID[3:0]). One CS interface of the external MCU can control at most 16 RF front-end chips. Therefore, the RF front-end chip can realize the control of one host and multiple slaves. The number of slave control is determined by the number of host chip selection interfaces ($16 \times$ number of host chip selection interfaces).

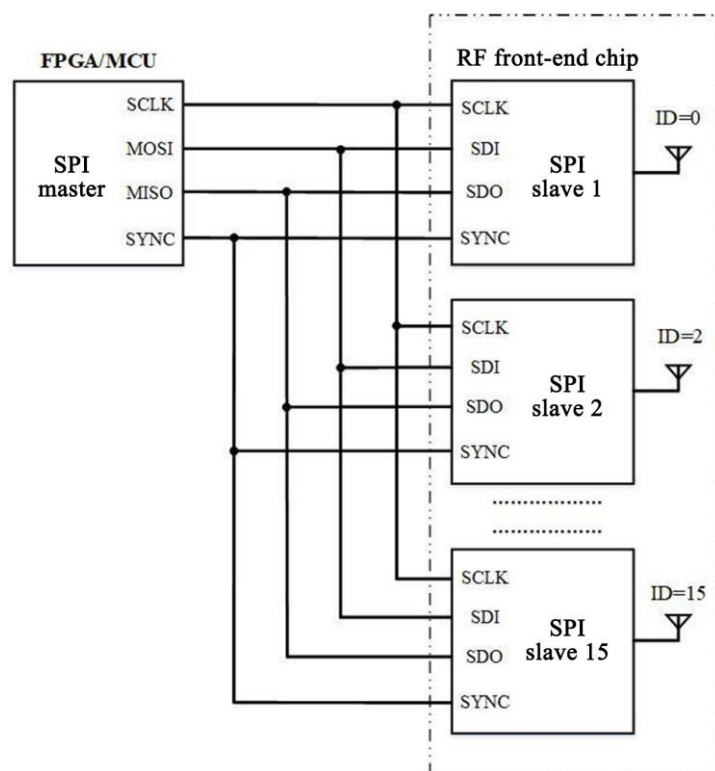


Figure 23 SPI Hardware Topology Diagram

Chip ID Configuration

The chip ID configuration only takes effect in the ID mode (MODE is 1), and the ID configuration methods of chips 0~15 are shown in the following table:

Table 6 Chip ID Configuration Table

D0~D3	Chip selection	D0~D3	Chip selection	D0~D3	Chip selection	D0~D3	Chip selection
0000	Chip 0	0100	Chip 4	1000	Chip 8	1100	Chip 12
0001	Chip 1	0101	Chip 5	1001	Chip 9	1101	Chip 13
0010	Chip 2	0110	Chip 6	1010	Chip 10	1110	Chip 14
0011	Chip 3	0111	Chip 7	1011	Chip 11	1111	Chip 15

Time Sequence

SPI communication protocol interfaces include SCLK, SYNC, SDI and SDO. Read and write operations can be carried out. SCLK is the system synchronization clock interface of SPI host and slave; SYNC is a low-level effective chip selection synchronization signal interface; SDI is the interface of host output (slave input); SDO is the interface of host input (slave output). Both input and output are in frames, and the length of each frame is fixed at 24 bits.

Write Time Sequence

As shown in Figure 24, when SYNC is lowered, the chip is selected. The serial input signal of SDI starts to be saved in the 24-bit buffer in the chip. Until all 24-bit signals are stored, when SYNC changes from low to high, data processing starts, corresponding registers are read and written, and SDO output is prepared. When SYNC is high, the rising edge of SCLK latches the data into the corresponding register.

For the amplitude phase control mode, after the rising edge of SCLK locks the data into the corresponding register, the falling edge of LDAC loads the amplitude and phase data to make the amplitude phase effective.

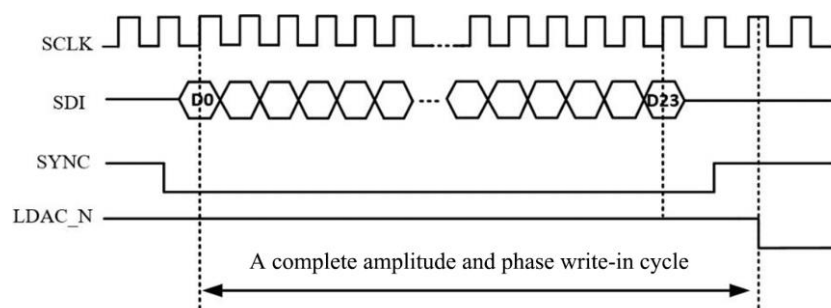


Figure 24 SPI Write Time Sequence

Note:

(1) To ensure the normal operation of the chip, after the current frame is sent and SYNC is pulled high, at least two SCLK rising edges should be separated before reading and writing the next frame;

(2) To ensure the normal operation of the chip, at least one SCLK cycle is required between the rising edge of SYNC and the falling edge of LDAC.

Read Time Sequence

The read time sequence exists only in the ID mode, that is, when the MODE is 1 (pull up to 3.3V). See Figure 25. On the falling edge of SYNC, SDI writes a 24-bit read command frame. On the rising edge of SYNC, the output content is put into SDO buffer in advance. On the next falling edge of SYNC, SDO starts to output the corresponding readback data. At the same time, the operation of writing a 24-bit read command frame to the next SDI will also take effect. If the next frame does not need to take effect, an invalid frame can be written with the correct ID No (for example, fill 1 after the ID). Note that the ID No. of the written frame must be correct, otherwise an error will occur in the output. At this time, the SDO state should not be focused when reading the command frame, because the current SDO state is only related to the previous frame.

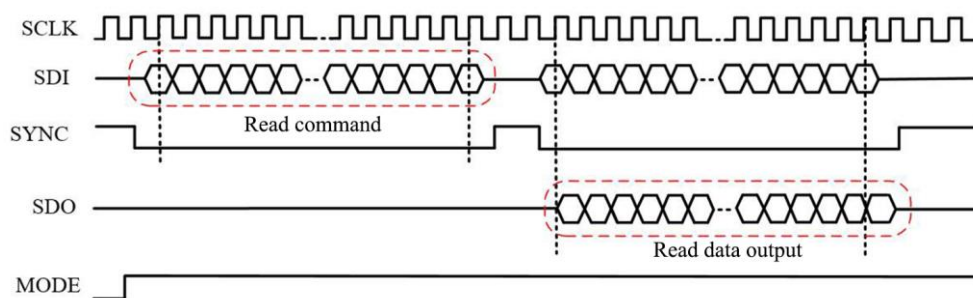


Figure 25 SPI Read Time Sequence

Continuous reading and writing work reflect the characteristics of an assembly line. Taking Figure 26 as an example, the readout data of reading command 1 will be given when the reading command 2 is written in.

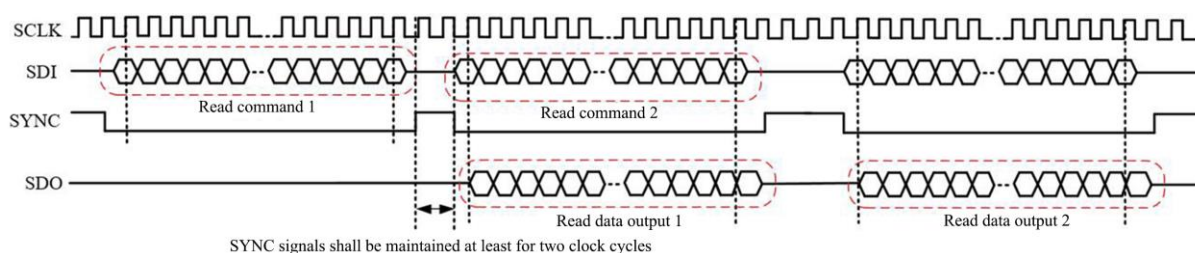


Figure 26 SPI Pipeline Schematic

Communication Protocol

Frame Composition

SPI communication is based on a fixed 24-bit frame. The overall frame structure is shown in Table 7. It can complete the functions of reading and writing channel power switch, amplitude and phase reading and writing, phase table reading and writing, on-chip control reading and writing, on-chip signal reading, pre-stored reading and writing, fast loading and pre-storing, etc.

The write command needs to be sent in a complete 24-bit frame. The read command only needs to send the frame to the end of the address, that is, the bold black dividing line position in the table. The following data part need not be sent. It can be filled with 0 or 1. SDO will return the complete readback frame in the next frame.

SPI has a simple recognition mechanism for error frames. When the structure of the input frame is greater than 24 bits, the frame is judged as an error frame and is directly discarded.

Table 7 Overall Frame Structure

Sending order	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
Channel switch	Chip ID			Read and write	Function selection			Enable	NULL											Channel number				
				Read 1 write 0	0	0	0	1	NULL											MSB			LSB	
Amplitude and phase	Chip ID			Read and write	Function selection			Channel selection			Amplitude						Phase							
				Read 1 write 0	0	0	1	MSB			LSB	MSB					LSB	MSB					LSB	
Calibration register	Chip ID			Read and write	Function selection					Register address						Register data								
				Read 1 write 0	0	1	0	0	0	MSB					LSB	MSB							LSB	
Signal register	Chip ID			Read and write	Function selection			Register address			Register data													
				Read only 1	0	1	0	1	MSB		LSB	MSB											LSB	
Control register	Chip ID			Read and write	Function selection			Register address								Register data								
				Read 1 write 0	0	1	1	MSB							LSB	MSB							LSB	

Register Function and Address Mapping

Channel Switch Register

Channel switch register can control the power supply switches of 4 Tx/Rx RF channels. 4 bits correspond to 4 channels, with the highest bit corresponding to channel 3 and the lowest bit corresponding to channel 0. High level on, low level off.

Table 8 Channel Switch Register Frame Structure

Chip ID				Read and write	Function selection			Channel enable	NULL											Channel number			
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
MSB		LSB		Read 1 write 0	0	0	0	1	NULL											MSB		LSB	

(1) Read/write configuration: During read/write configuration, D4 is 1 for read operation and D4 is 0 for write operation.

(2) Function selection configuration: D5~D7 are function selection bits, and D5~D7 must be [000] when configuring the channel switch register.

(3) Channel enable configuration: D8 is the channel enable bit. To configure the channel switch register, D8 must be [1].

(4) Idle bit configuration: D9~D19 are idle bits that can be filled with 0 or 1.

(5) Channel number configuration: D20~D23 are channel number control bits, and the truth value table is as follows.

Table 9 Channel No. Configuration Truth Table

D20	D21	D22	D23	Channel Status
0	0	0	0	Channel fully off
0	0	0	1	Channel 0 on
0	0	1	0	Channel 1 on
0	1	0	0	Channel 2 on
1	0	0	0	Channel 3 on
1	1	1	1	Channel fully on

Example: in ID mode (MODE is 1), carry out power supply write operation to channel 1 of chip 1(ID is 1).

The corresponding 24-bit frame should be configured as: D0~D23[0001_0_000_1_1111111111_0010].

Amplitude and Phase Register

The amplitude register can configure the amplitude and phase of the designated channel. In which the amplitude and phase configurations each occupy 6 bits.

Table 10 Frame Structure of Amplitude and Phase Register

Chip ID				Read and write	Function selection			Channel enable	Receiving and transmitting selection	Channel selection		Amplitude						Phase					
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
MSB		LSB		Read 1 write 0	0	0	1	0	T0R1	MSB	LSB	MSB						MSB					

(1) Transceiver selection configuration: D9 is the transceiver selection control bit, D9 is 0 to configure the transmit state, and D9 is 1 to configure the receive state.

(2) Channel selection configuration: D10~D11 are channel selection control bits, and the truth table is as follows.

Table 11 Amplitude Phase Register Channel Selection Truth Table

Channel number	Channel 0	Channel 1	Channel 2	Channel 3
D10	0	0	1	1

Channel number	Channel 0	Channel 1	Channel 2	Channel 3
D11	0	1	0	1

(3) Amplitude configuration: The amplitude control occupies 6 bits in total. (D12 is an idle bit). The maximum amplitude attenuation can reach 15.5 dB.

Table 12 Amplitude Configuration Truth Table

D13~D17	Amplitude modulation value (dB)	D13~D17	Amplitude modulation value (dB)	D13~D17	Amplitude modulation value (dB)	D13~D17	Amplitude modulation value (dB)
00_0000	0	00_1000	-4	01_0000	-8	01_1000	-12
00_0001	-0.5	00_1001	-4.5	01_0001	-8.5	01_1001	-12.5
00_0010	-1	00_1010	-5	01_0010	-9	01_1010	-13
00_0011	-1.5	00_1011	-5.5	01_0011	-9.5	01_1011	-13.5
00_0100	-2	00_1100	-6	01_0100	-10	01_1100	-14
00_0101	-2.5	00_1101	-6.5	01_0101	-10.5	01_1101	-14.5
00_0110	-3	00_1110	-7	01_0110	-11	01_1110	-15
00_0111	-3.5	00_1111	-7.5	01_0111	-11.5	01_1111	-15.5

(4) Phase configuration: Phase control occupies 6 bits, and D18 (the highest bit) - D23 (the lowest bit) corresponds to 180° - 5.625° respectively.

Table 13 Phase Configuration Truth Table

D18~D23	Phase modulation value ($^\circ$)	D18~D23	Phase modulation value ($^\circ$)	D18~D23	Phase modulation value ($^\circ$)	D18~D23	Phase modulation value ($^\circ$)
00_0000	0	01_0000	-90	10_0000	-180	11_0000	-270
00_0001	-5.625	01_0001	-95.625	10_0001	-185.625	11_0001	-275.625
00_0010	-11.25	01_0010	-101.25	10_0010	-191.25	11_0010	-281.25
00_0011	-16.875	01_0011	-106.875	10_0011	-196.875	11_0011	-286.875
00_0100	-22.5	01_0100	-112.5	10_0100	-202.5	11_0100	-292.5
00_0101	-28.125	01_0101	-118.125	10_0101	-208.125	11_0101	-298.125
00_0110	-33.75	01_0110	-123.75	10_0110	-213.75	11_0110	-303.75
00_0111	-39.375	01_0111	-129.375	10_0111	-219.375	11_0111	-309.375
00_1000	-45	01_1000	-135	10_1000	-225	11_1000	-315
00_1001	-50.625	01_1001	-140.625	10_1001	-230.625	11_1001	-320.625
00_1010	-56.25	01_1010	-146.25	10_1010	-236.25	11_1010	-326.25
00_1011	-61.875	01_1011	-151.875	10_1011	-241.875	11_1011	-331.875
00_1100	-67.5	01_1100	-157.5	10_1100	-247.5	11_1100	-337.5
00_1101	-73.125	01_1101	-163.125	10_1101	-253.125	11_1101	-343.125
00_1110	-78.75	01_1110	-168.75	10_1110	-258.75	11_1110	-348.75
00_1111	-84.375	01_1111	-174.375	10_1111	-264.375	11_1111	-354.375

Control Register

This chip has a total of 256 (actually 27 used) 8-bit control registers. This type of register has no channel enable configuration, and the channel information is contained in the register address. The address and function mapping of the control register are shown in Table 14.

Table 14 Control Register Frame Structure

Chip ID				Read and write	Function selection			Register address								Register data							
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
MSB				Read 1 write 0	0	1	1	MSB								LSB							

Table 15 Control Register Address Mapping

S/N	Function	Register address	Register value	Description
Chip bias				
1	CH1 TX PA bias	0000_0000	0010_0000	Eight-bit bias current DAC, with the [5:0] bits valid, the upper two bits as 00 by default , and the lowest bit as 0 by default
2	CH1 TX DA bias	0000_0001	0010_0000	
3	CH1 TX AMP bias	0000_0010	0010_0000	
4	CH1 RX LNA bias	0000_0011	0010_0110	
5	CH1 RX AMP bias	0000_0100	0001_1110	
6	CH2 TX PA bias	0000_1000	0010_0000	
7	CH2 TX DA bias	0000_1001	0010_0000	
8	CH2 TX AMP bias	0000_1010	0010_0000	
9	CH2 RX LNA bias	0000_1011	0010_0110	
10	CH2 RX AMP bias	0000_1100	0001_1110	
11	CH3 TX PA bias	0001_0000	0010_0000	
12	CH3 TX DA bias	0001_0001	0010_0000	
13	CH3 TX AMP bias	0001_0010	0010_0000	
14	CH3 RX LNA bias	0001_0011	0010_0110	
15	CH3 RX AMP bias	0001_0100	0001_1110	
16	CH4 TX PA bias	0001_1000	0010_0000	
17	CH4TX DA bias	0001_1001	0010_0000	
18	CH4 TX AMP Bias	0001_1010	0010_0000	
19	CH4 RX LNA bias	0001_1011	0010_0110	
20	CH4 RX AMP bias	0001_1100	0001_1110	
2	TX PA temperature coefficient	CTRL247[7:0]	0101_1010	
3	TX AMP temperature coefficient	CTRL250[7:0]	0101_1010	
4	RXLNA amplifier	CTRL251[7:0]	0101_1010	
5	RX AMP bias	CTRL254[7:0]	0101_1010	

Note: The address of the control register is controlled by a total of 8 bits of data from D8 to D15, and the decimal should be converted into binary in the SPI read-write communication frames. For example, CTRL38[7:0] shall be written as 0010_0110.

Example: in ID mode (MODE is 1), carry out write operation to the TX power amplifier bias of channel 0 of chip 1 (ID is 1).

The corresponding 24-bit frame should be configured as: D0-D23[0001_0_011_00000001_00100000].

Temperature Coefficient Control Register

Table 16 Temperature Coefficient Control Register Address Mapping

S/N	Function	Register address	Register initial value	Description
Temperature compensation coefficient				
1	TX DA temperature coefficient	CTRL246[7:0]	0101_1010	Temperature compensation is controlled by registers P and Z respectively, with the first four bits by P and the last four bits by Z, and the value of register Z is the bitwise inversion of P; therefore, there are 16 temperature compensation states. 1111_0000 corresponds to the maximum temperature compensation state and 0000_1111 corresponds to the minimum temperature compensation state.
2	TX PA temperature coefficient	CTRL247[7:0]	0101_1010	
3	TX AMP temperature coefficient	CTRL250[7:0]	0101_1010	
4	RXLNA amplifier	CTRL251[7:0]	0101_1010	
5	RX AMP bias	CTRL254[7:0]	0101_1010	

It is recommended that the temperature coefficient be configured according to the initial values in the table. Refer to Table 14 for the 24-bit frame structure.

High-Low Speed Switching Control Register

Table 17 High-Low Speed Switching Control Register Address Mapping

Function	Register address	Initial Value	Description
High-low speed switching			
High-low speed switching control	CTRL242[3:0]	0000	From the highest bit to the lowest bit, it corresponds to the high-low speed switching control of CH3, CH2, CH1, and CH0. When the value is 0, it is in high-speed switching with extra power consumption because the IDAC module is always turned on at this time; when the value is 1, it is in low-speed switching without extra power consumption.

This register is used to control the switching time of the receiving and transmitting mode. In the high-speed switching state, the receiving and transmitting switching time can be 50ns, and in the low-speed switching state, the receiving and transmitting switching time can be 1μs. Refer to Table 14 for the 24-bit frame structure.

Signal Register

The signal register is in read-only mode. The address and function mapping are shown in Table 18, and 10 bits (SIG[10] and SIG[11] are invalid bits) of ADC output temperature and power data can be read out.

Table 18 Signal Register Frame Structure

Chip ID				Read and write	Function selection			Channel enable	Register address			Register data											
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
MSB		LSB		Read only 1	0	1	0	1	MSB		LSB	MSB										LSB	

Table 19 Signal Register Address Mapping

S/N	Function	Register address	Readback value	Description
1	ADC input data	SIG00[9:0]	N/A	

Note: Before reading the 10-bit ADC output temperature in the signal register SIG00, the user needs to write the 24-bit frame in the control registers CTRL16, CTRL244 and CTRL245. Then perform the read-only operation of the signal register SIG00. (see "temperature readback operation" for details)

Temperature Readback

ADC Control Register

Table 20 Temperature Sensor and ADC Control Register Address Mapping

S/N	Function	Register address	Register initial value	Description
ADC & Power detection & Temperature sensor				
1	ADC input selection	CTRL244[4:0]	0_0000	Initial value 0_0000 of register: Input temperature;
2	ADC clock enable	CTRL16[3]	1	1 ON/0 OFF
3	ADC clock frequency division factor	CTRL16[2:0]	100	000-111: divided to 23 to 210 times
4	Temperature detection & calibration RES[5:0]	CTRL245[5:0]	10_0000	1 bit corresponds to a temperature change of 0.4 degrees

Temperature Readback Operation

Step 1: Conduct reset operation of RST_IN after the chips are powered on;

The advantages of this design include reduced control board costs and the use of low-cost FR4 materials in layers without RF feed networks.

Step 2: As shown in Table 20, write the 24-bit frames of CTRL244, CTRL16 and CTRL245 control registers in turn;

Step 3: Wait for at least 2400 SCLK clock cycles;

Step 4: Write 24-bit frames used to read back the SIG00 signal register;

Step 5: Read back the return value of SIG00 signal register to obtain the temperature.

Temperature calculation: $\text{Temperature} = 249 - 0.364 * \text{Code}_{(D)}$

Example: When temperature check calibration RES[5:0] = 10_0000 (default), the ADC output corresponding to 27°C is 10_0110_0100 (610); when temperature check calibration RES = 11_1111, the ADC output corresponding to 27°C is 10_1000_0100 (644);

Table 21 Correspondence between Temperature and ADC Output

Temperature [°C]	-40	-20	0	20	27	40	60	80	100
ADC Output (D)	794	739	684	629	610	574	519	464	408

There is an error (maximum $\pm 12^\circ\text{C}$) in the corresponding temperature of the same ADC reading, and the relative difference is unchanged. It can be compensated by temperature detection & calibration RES[5:0].

Chip SPI Configuration Steps

Write Operation of Chip

Step 1: Conduct reset operation of RST after the chips is powered on;

Step 2: Write 24-bit frames corresponding to different control registers in turn, and configure chip bias, temperature compensation, high-low speed switching, etc.

Step 3: Write a 24-bit frame corresponding to the channel switch register;

Step 4: Write a 24-bit frame corresponding to the T/R amplitude and phase control register;

Read Operation of Chip

Step 1: Conduct reset operation of RST after the chips is powered on;

Step 2: Write a 24-bit frame corresponding to each register used for readback.

Step 3: Read back the return value of each register.